LESSON PLAN · SCRATCH (GR 3–5) INTERMEDIATE: STORY, QUIZ & MUSIC STUDIO

Lesson 8: Random Questions, Score and Timer

Module 3 | 45 minutes | Random, variables and timing | Grades 3–5 · ages 8–10

Objective: Generate random quiz questions, keep score in a variable, and add a time limit with the timer.

Success criteria

- I can make random questions with variables.
- I can keep score correctly.
- I can add a time limit with the timer.

Key vocabulary

- **random**: Chosen by chance.
- **score**: A variable counting correct answers.
- **timer**: Counts seconds since it was reset.
- **increment**: Add 1 to a variable.

You will need

- Projector with Kitabash (lesson video, Code Lab)
- Devices: 1 per learner or pair
- Printed worksheet (optional)

Timed lesson plan (45 min)

Time	Stage	What happens
5 min	**Warm-up (unplugged)**	Dice quiz (5 min): roll two dice, the class multiplies; keep a tally on the board (score).
5 min	**Watch**	Play the lesson video with captions; pause at each heading and ask a question.
8 min	**Build along**	A random times-table quiz: learners follow the steps.
14 min	**Perform**	Code Lab: random quiz, Reset the score
9 min	**Probe**	Code Lab: time limit, Trace the score, activity
4 min	**Wrap-up**	Exit ticket: Why must the random numbers be stored in variables?

Unplugged warm-up

Dice quiz (5 min): roll two dice, the class multiplies; keep a tally on the board (score).

Teaching the idea

Random numbers stored in variables make each question unique; the expected answer is computed with operators. A score variable is initialised and incremented inside the correct branch. The timer reporter counts seconds since reset, enabling time limits.

Analogy: Rolling two dice for every question: the quiz master doesn't know the question in advance either!

Live demo: Run the random quiz three times to show different questions. Then show what happens if you forget "set score to 0".

Questions to ask

Ask	Listen for
Why store the random numbers in variables?	So we can use the same numbers in the question and the check.
Where does "change score by 1" go?	Inside the if (correct) branch.
How can the game end after 30 seconds?	reset timer and wait until timer > 30.

Differentiation

Support
Pre-make the variables and the question join; learners add the check and score.

Stretch
Add levels: after 3 correct answers, numbers go up to 20. Or give a bonus point for fast answers using the timer.

Common misconceptions

Misconception	How to fix it
pick random in the question and again in the check gives the same number	Each pick random gives a new number; store it once in a variable.
The timer resets itself	Use "reset timer" at the start.
Score goes in the else branch	Only correct answers add points.

Assessment

Use the Code Lab results (auto-checked), the quick checks and the Probe answers. Written answers arrive in your review queue.

Exit ticket
Why must the random numbers be stored in variables?

Homework
Play a times-table quiz with someone at home: who gets the higher score in 1 minute?

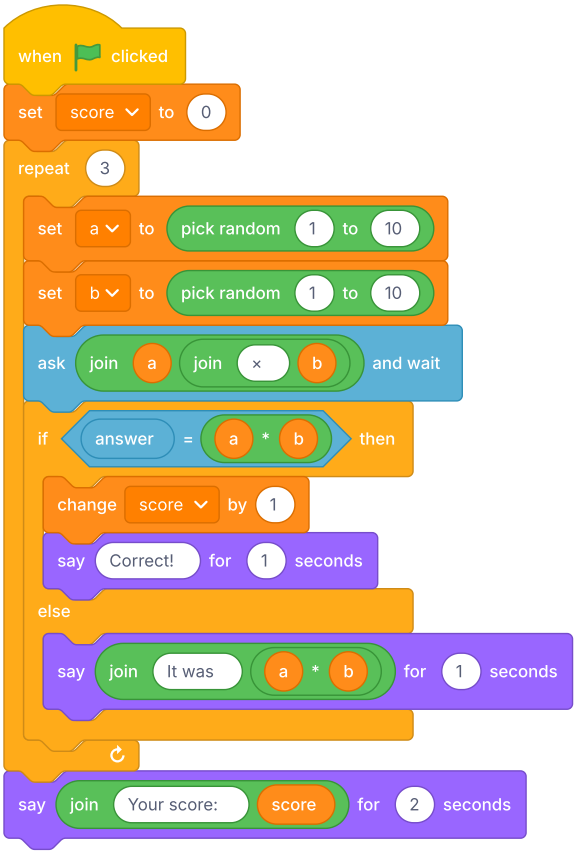
Answer key — Kitabash activities

1. Code Lab: random quiz (perform · code lab)

Goals checked: Random numbers · Asks 3 questions · Checks with × · Score starts at 0

One solution:

Host



2. Reset the score (perform · multiple choice)

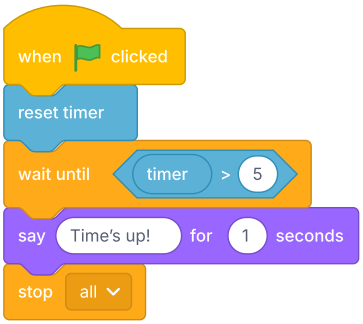
Answer: So each new game starts from zero

3. Code Lab: time limit (probe · code lab)

Goals checked: Resets the timer · Says “Time’s up!” after 5 seconds · Stops the game

One solution:

Host



4. Trace the score (probe · multiple choice)

Answer: 2

5. Explain your thinking (probe · free response)

Look for: random, timer, bigger. **Example:** I would change pick random 1 to 10 into 1 to 20 so the numbers are bigger. I would also reset the timer and only give 5 seconds per question.

Answer key — worksheet

1. **Key words:** random = Chosen by chance. · score = A variable counting correct answers. · timer = Counts seconds since it was reset. · increment = Add 1 to a variable.
2. **Plan your code:** Goals: Random numbers; Asks 3 questions; Checks with ×; Score starts at 0.
3. **Reset the score:** A. So each new game starts from zero — Variables keep their value between runs unless you reset them.
4. **Plan your code:** Goals: Resets the timer; Says “Time’s up!” after 5 seconds; Stops the game.
5. **Trace the score:** A — Score only goes up inside the if, when the answer is correct: twice.

Kitabash original curriculum. Scratch is a project of the Scratch Foundation in collaboration with the MIT Media Lab; this course is not affiliated with or endorsed by them.